

BITP 3513 ADVANCED DATABASE PROGRAMMING

INDIVIDUAL ASSIGNMENT 2-a (10%)

Read the case study below and answer the following question.

Case Study: Sales Performance Monitoring System

A small retail store records product sales information in a database. The store owner wants an Oracle APEX application to analyse sales performance, automate notifications, send emails, and transfer the application data to another environment.

Data Dictionary of Product_Sales		
Attribute	Data Type	Description
SALE_ID	NUMBER	Unique sales transaction ID (Primary Key)
PRODUCT_NAME	VARCHAR2(100)	Product name
CATEGORY	VARCHAR2(50)	Product category (Electronics, Stationery, Food, etc.)
QUANTITY_SOLD	NUMBER	Quantity sold
UNIT_PRICE	NUMBER(10,2)	Price per unit
TOTAL_SALES	NUMBER(10,2)	Total sales amount
SALE_DATE	DATE	Date of sale
BRANCH_NAME	VARCHAR2(50)	Branch where the sale occurred
STATUS	VARCHAR2(20)	Status updated by Automation (e.g., Normal, Top Seller)

Product_Sales.sql
<pre>CREATE TABLE PRODUCT_SALES (SALE_ID NUMBER PRIMARY KEY, PRODUCT_NAME VARCHAR2(100), CATEGORY VARCHAR2(50), QUANTITY_SOLD NUMBER, UNIT_PRICE NUMBER(10,2), TOTAL_SALES NUMBER(10,2), SALE_DATE DATE, BRANCH_NAME VARCHAR2(50), STATUS VARCHAR2(20));</pre>

Based on case study, the information of data dictionary and SQL code were provided. There are FOUR (4) tasks. Answer all questions by follow the step given for each task. Provide short explanation for each screenshot.

Task 1: Create Charts

1. Create a table named PRODUCT_SALES.
2. Insert at least 30 sales records.
3. Create an Interactive Report to display the sales data.
4. Create ONE (1) suitable charts to visualise the sales information.
5. Provide a screenshot of the charts and table of sales reports.

Task 2: Create an Automation

1. Create an Automation that runs on the PRODUCT_SALES table.
2. Configure the Automation to update a status field or generate a summary record.
3. Execute the Automation.
4. Review the Automation Execution Log.
5. Provide a screenshot of the Automation configuration and Automation Execution Log.

Task 3: Create a Send Email Process

1. Create a page that allows users to submit sales information.
2. Configure a Send Email process to send a notification after form submission.
3. Test the email process.
4. Provide a screenshot of the Form Page, Send Email process and email queue or received email.

Task 4: Migrate Database Objects and Data

1. Export the PRODUCT_SALES table and its data.
2. Import the exported file into another workspace or schema.
3. Verify that the table and records are successfully imported.
4. Provide a screenshot of the Export process, Import process and showing the imported records.

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Answer:

TASK 1 : Create Charts

a. Create table PRODUCT_SALES.

The screenshot shows the Oracle APEX SQL Workshop interface. The SQL Commands pane contains the following SQL statement:

```

1 CREATE TABLE PRODUCT_SALES (
2   SALE_ID NUMBER(10) PRIMARY KEY,
3   PRODUCT_NAME VARCHAR2(100),
4   CATEGORY VARCHAR2(50),
5   QUANTITY_SOLD NUMBER,
6   UNIT_PRICE NUMBER(8,2),
7   TOTAL_SALES NUMBER(10,2),
8   SALE_DATE DATE,
9   BRANCH_NAME VARCHAR2(50),
10  STATUS VARCHAR2(20)
11 );

```

The Results pane shows the message: "Table created." and the execution time: "0.02 seconds".

The PRODUCT_SALES table was created in Oracle Database based on the given data dictionary. The table stores sales transaction information including product details, sales amount, and status.

b. Insert Value

The screenshot shows the Oracle APEX SQL Workshop interface. The SQL Commands pane contains the following SQL statement:

```

1 BEGIN
2   INSERT INTO PRODUCT_SALES VALUES (1,'Laptop','Electronics',2,2500,5000,DATE '2025-01-05','Melaka','Normal');
3   INSERT INTO PRODUCT_SALES VALUES (2,'Mouse','Electronics',10,15,150,DATE '2025-01-06','Melaka','Normal');
4   INSERT INTO PRODUCT_SALES VALUES (3,'Keyboard','Electronics',15,40,600,DATE '2025-01-07','KL','Normal');
5   INSERT INTO PRODUCT_SALES VALUES (4,'Notebook','Stationery',100,1,50,DATE '2025-01-08','KL','Normal');
6   INSERT INTO PRODUCT_SALES VALUES (5,'Pen','Stationery',50,1,50,DATE '2025-01-09','Johor','Normal');
7   INSERT INTO PRODUCT_SALES VALUES (6,'Printer','Electronics',1,450,450,DATE '2025-01-10','Melaka','Normal');
8   INSERT INTO PRODUCT_SALES VALUES (7,'Bread','Food',20,5,100,DATE '2025-01-11','Johor','Normal');
9   INSERT INTO PRODUCT_SALES VALUES (8,'Milk','Food',15,6,90,DATE '2025-01-12','KL','Normal');
10  INSERT INTO PRODUCT_SALES VALUES (9,'Chair','Furniture',3,120,360,DATE '2025-01-13','Melaka','Normal');
11  INSERT INTO PRODUCT_SALES VALUES (10,'Table','Furniture',2,200,600,DATE '2025-01-14','KL','Normal');
12
13  INSERT INTO PRODUCT_SALES VALUES (11,'Monitor','Electronics',4,600,2400,DATE '2025-02-01','Johor','Normal');
14  INSERT INTO PRODUCT_SALES VALUES (12,'USB Cable','Electronics',25,10,250,DATE '2025-02-02','Melaka','Normal');
15  INSERT INTO PRODUCT_SALES VALUES (13,'Pencil','Stationery',40,1,40,DATE '2025-02-03','KL','Normal');
16  INSERT INTO PRODUCT_SALES VALUES (14,'File','Stationery',20,5,100,DATE '2025-02-04','Johor','Normal');
17  INSERT INTO PRODUCT_SALES VALUES (15,'Rice','Food',10,25,250,DATE '2025-02-05','Melaka','Normal');
18  INSERT INTO PRODUCT_SALES VALUES (16,'Eggs','Food',12,12,144,DATE '2025-02-06','KL','Normal');
19  INSERT INTO PRODUCT_SALES VALUES (17,'Sofa','Furniture',1,900,900,DATE '2025-02-07','Johor','Normal');

```

The Results pane shows the message: "1 row(s) inserted." and the execution time: "0.05 seconds".

A total of more than 30 sales records were inserted into the PRODUCT_SALES table to provide sufficient data for analysis and reporting.

Result :

The screenshot shows the APEX SQL Workshop interface with the 'PRODUCT_SALES' table selected. The table contains 15 rows of sales data. The columns are: SALE_ID, PRODUCT_NAME, CATEGORY, QUANTITY_SOLD, UNIT_PRICE, TOTAL_SALES, SALE_DATE, BRANCH_NAME, and STATUS.

SALE_ID	PRODUCT_NAME	CATEGORY	QUANTITY_SOLD	UNIT_PRICE	TOTAL_SALES	SALE_DATE	BRANCH_NAME	STATUS
1	Laptop	Electronics	2	2500	5000	05-Jan-2026	Melaka	Normal
2	Mouse	Electronics	10	35	350	06-Jan-2026	Melaka	Normal
3	Keyboard	Electronics	5	80	400	07-Jan-2026	KL	Normal
4	Notebook	Stationery	30	3	90	08-Jan-2026	KL	Normal
5	Pen	Stationery	50	1.5	75	09-Jan-2026	Johor	Normal
6	Printer	Electronics	1	450	450	10-Jan-2026	Melaka	Normal
7	Bread	Food	20	4	80	11-Jan-2026	Johor	Normal
8	Milk	Food	15	6	90	12-Jan-2026	KL	Normal
9	Chair	Furniture	3	120	360	13-Jan-2026	Melaka	Normal
10	Table	Furniture	2	300	600	14-Jan-2026	KL	Normal
11	Monitor	Electronics	4	600	2400	01-Feb-2026	Johor	Normal
12	USB Cable	Electronics	25	10	250	02-Feb-2026	Melaka	Normal
13	Pencil	Stationery	40	1	40	03-Feb-2026	KL	Normal
14	File	Stationery	20	5	100	04-Feb-2026	Johor	Normal
15	Rice	Food	10	25	250	05-Feb-2026	Melaka	Normal

c. Create Application + Interactive Report

The screenshot shows the 'Create an Application' wizard in APEX. The application name is 'Sales Performance Monitoring System' and the appearance is 'Vita, Top Menu'. The 'Pages' section shows a 'Home' page. The 'Features' section has several options checked, including 'Install Progressive Web App', 'Activity Reporting', 'Theme Style Selection', 'About Page', 'Configuration Options', 'Access Control', and 'Feedback'.

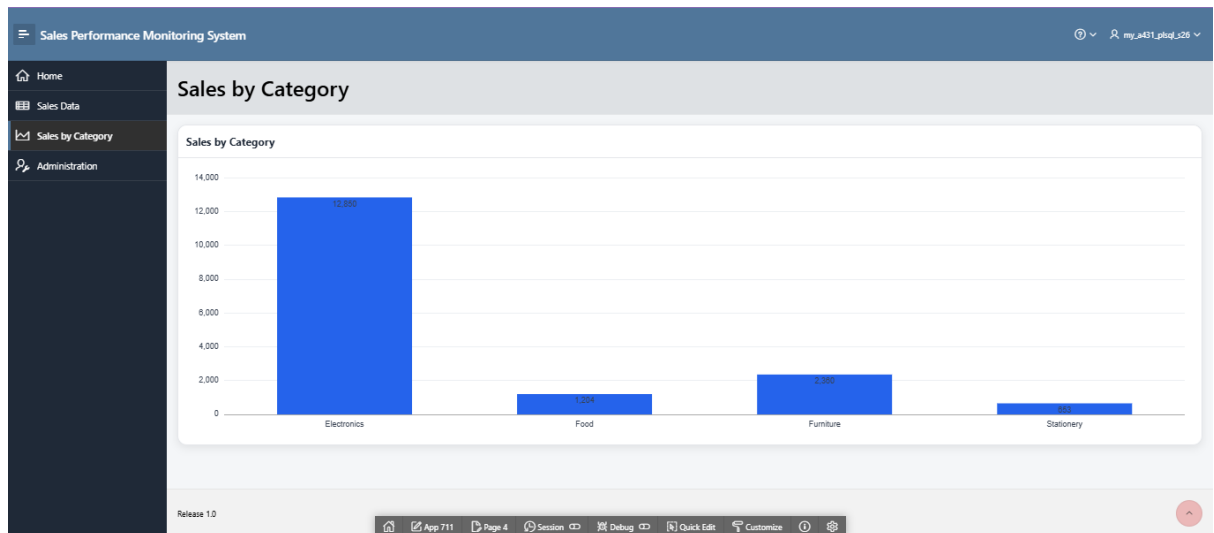
I. Interactive Report – sales product

The screenshot shows the 'Sales Performance Monitoring System' application. The 'Sales Data' report is displayed, showing a table of sales records. The table has columns for Product Name, Category, Quantity Sold, Unit Price, Total Sales, Sale Date, Branch Name, and Status. The data is sorted by Sale Date in descending order.

Product Name	Category	Quantity Sold	Unit Price	Total Sales	Sale Date	Branch Name	Status
Laptop	Electronics	2	2500	5000	1/5/2026	Melaka	Normal
Mouse	Electronics	10	35	350	1/6/2026	Melaka	Normal
Keyboard	Electronics	5	80	400	1/7/2026	KL	Normal
Notebook	Stationery	30	3	90	1/8/2026	KL	Normal
Pen	Stationery	50	1.5	75	1/9/2026	Johor	Normal
Printer	Electronics	1	450	450	1/10/2026	Melaka	Normal
Bread	Food	20	4	80	1/11/2026	Johor	Normal
Milk	Food	15	6	90	1/12/2026	KL	Normal
Chair	Furniture	3	120	360	1/13/2026	Melaka	Normal
Table	Furniture	2	300	600	1/14/2026	KL	Normal
Monitor	Electronics	4	600	2400	2/1/2026	Johor	Normal
USB Cable	Electronics	25	10	250	2/2/2026	Melaka	Normal
Pencil	Stationery	40	1	40	2/3/2026	KL	Normal
File	Stationery	20	5	100	2/4/2026	Johor	Normal
Rice	Food	10	25	250	2/5/2026	Melaka	Normal
Eggs	Food	10	25	250	2/6/2026	KL	Normal

An Interactive Report was created to display all sales records stored in the PRODUCT_SALES table. Users can search, sort, and filter the sales data easily.

ii. Chart



A bar chart was created to visualise total sales by product category. The chart helps users identify which category generates the highest sales.

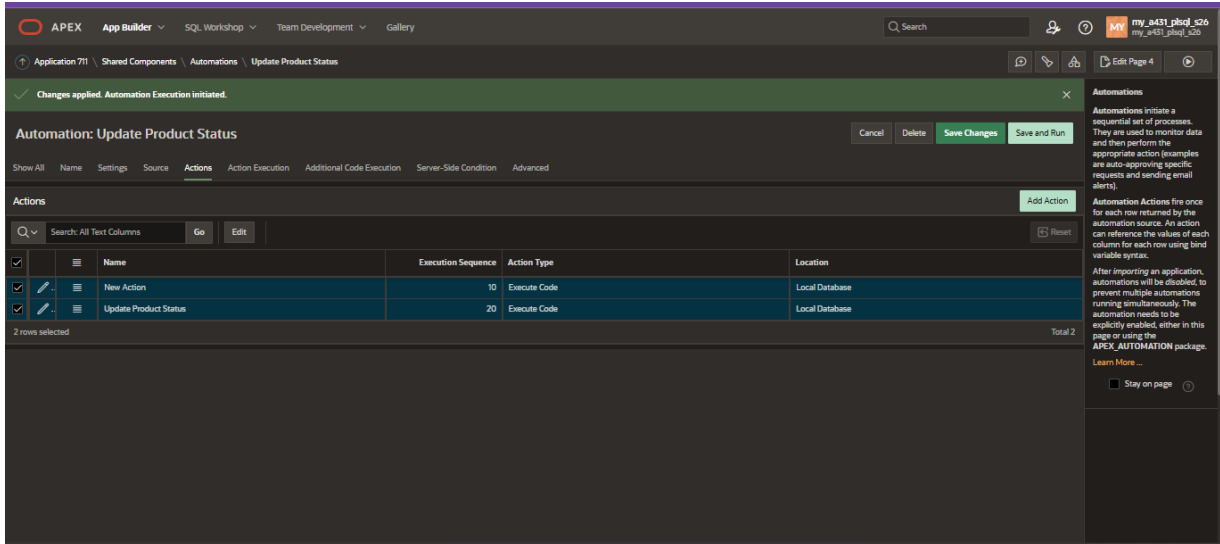
Summary task 1 :

In this task, the PRODUCT SALES table was created and populated with sales records. An Interactive Report and a bar chart were developed in Oracle APEX to display and analyse sales performance by category.

Task 2 : Create Automation

i. Automation Configuration

Step : app builder – share component – automations – create automation – create action



Automation: Update Product Status

Cancel Delete Save Changes Save and Run

Show All Name Settings Source Actions Action Execution Additional Code Execution Server-Side Condition Advanced

Actions

Search: All Text Columns Go Edit

Name	Execution Sequence	Action Type	Location
New Action	10	Execute Code	Local Database
Update Product Status	20	Execute Code	Local Database
Total 2			

2 rows selected

Automation Actions fire once for each row returned by the automation source. An action can reference the value of each column for each row using bind variable syntax.

After importing an application, automations will be disabled to prevent multiple automations running simultaneously. The automation needs to be explicitly enabled, either in this page or using the APEX AUTOMATION package.

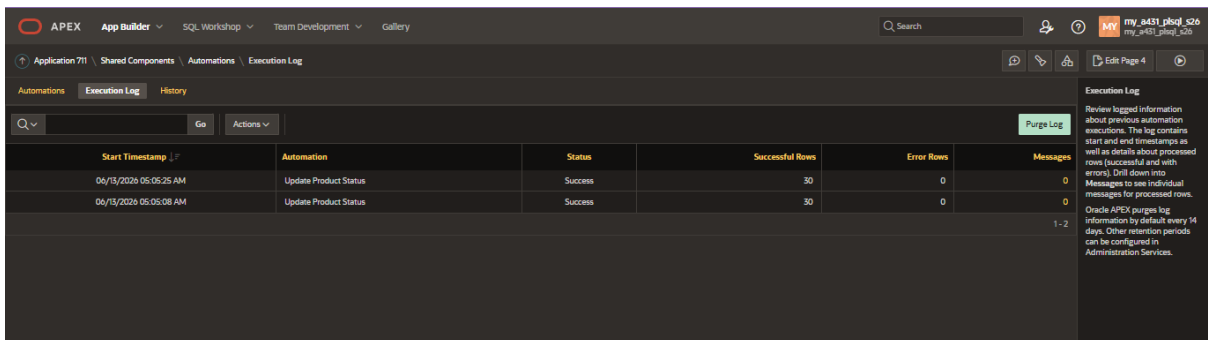
Learn More...

Stay on page

An automation was configured on the PRODUCT_SALES table to automatically update the STATUS field based on the total sales amount.

ii. Automation Execution Log

Step : share component – automation – execution log



Automation Execution Log

Search: All Text Columns Go Actions

Start Timestamp	Automation	Status	Successful Rows	Error Rows	Messages
06/13/2026 05:05:25 AM	Update Product Status	Success	30	0	0
06/13/2026 05:05:08 AM	Update Product Status	Success	30	0	0

1 - 2

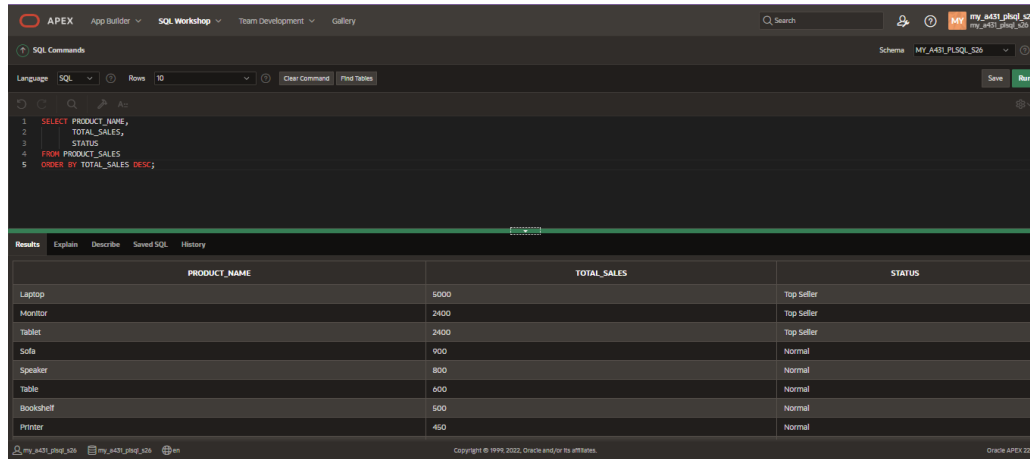
Execution Log

Review logged information about previous automation executions. The log contains start and end timestamps as well as details about processed rows (successful and with errors). Drill down into Messages to view individual messages for processed rows.

Oracle APEX purges log information by default every 14 days. Other retention periods can be configured in Administration Services.

The Automation Execution Log shows that the automation was executed successfully and processed the sales records.

Check data



The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user information 'my_a431_phaj_s26' are on the right. The 'SQL Commands' tab is active, displaying a SQL query:

```
1 SELECT PRODUCT_NAME,  
2       TOTAL_SALES,  
3       STATUS  
4 FROM PRODUCT_SALES  
5 ORDER BY TOTAL_SALES DESC;
```

 Below the query editor, the 'Results' tab shows a table with 10 rows. The table has three columns: 'PRODUCT_NAME', 'TOTAL_SALES', and 'STATUS'. The data is sorted by 'TOTAL_SALES' in descending order. The first three products (Laptop, Monitor, Tablet) are 'Top Seller', while the remaining seven (Tablet, Sofa, Speaker, Table, Bookshelf, Printer) are 'Normal'.

PRODUCT_NAME	TOTAL_SALES	STATUS
Laptop	5000	Top Seller
Monitor	2400	Top Seller
Tablet	2400	Top Seller
Sofa	900	Normal
Speaker	800	Normal
Table	600	Normal
Bookshelf	500	Normal
Printer	450	Normal

The updated sales records were verified to ensure that the STATUS field was correctly updated by the automation.

Summary Task 2 :

In this task, an automation was configured to automatically update the STATUS field in the PRODUCT_SALES table. The automation was executed successfully, and the execution log was reviewed to verify the results.

Task 3 : Create a Send Email Process

a. Create Page Form

The screenshot shows a web application titled 'Sales Performance Monitoring System'. On the left is a sidebar menu with options: Home, Sales Data, Sales by Category, Add New Sales (selected), and Administration. The main content area is titled 'Add New Sales' and contains a form with the following fields: Product Name, Category, Quantity Sold, Unit Price, Total Sales, Sale Date, Branch Name, and Status. At the bottom of the form are 'Cancel' and 'Create' buttons. The bottom of the page features a navigation bar with links to App 711, Page 3, Session, Debug, Quick Edit, Customize, and a settings icon.

A form page named Add New Sales was created to allow users to submit new sales information into the PRODUCT_SALES table.

i. Create auto sequenced ID

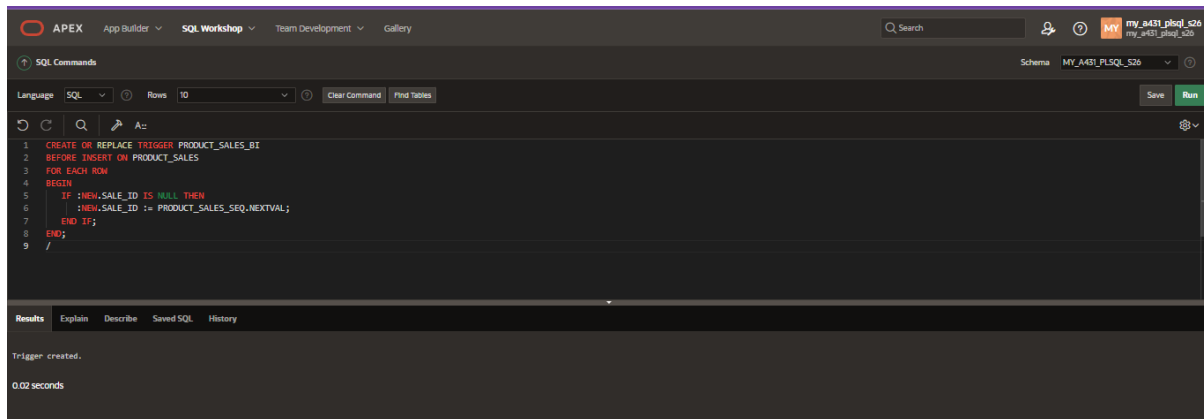
A database sequence was created to automatically generate unique SALE_ID values for each new sales record.

The screenshot shows the APEX SQL Workshop interface. The 'SQL Commands' tab is active, displaying the following SQL code:

```
1 CREATE SEQUENCE PRODUCT_SALES_SEQ
2 START WITH 31
3 INCREMENT BY 1;
```

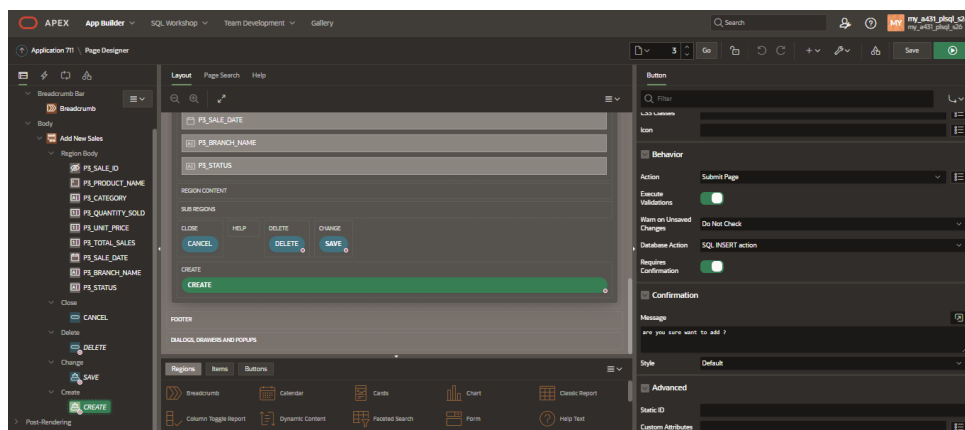
The 'Results' tab at the bottom shows the message 'Sequence created.' and the execution time '0.01 seconds'.

ii. Create trigger

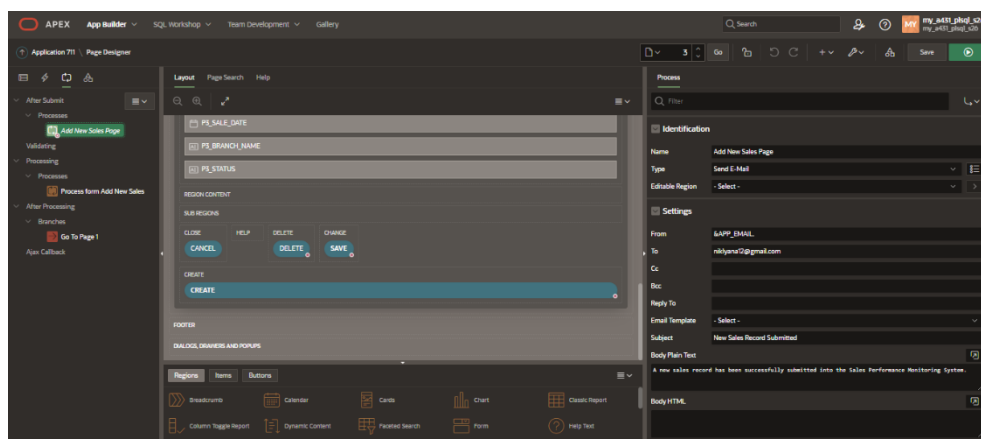


A database trigger was created to assign the next sequence value to SALE_ID before a new record is inserted.

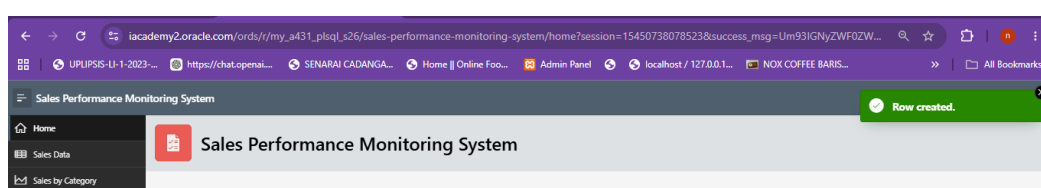
iii. Setting Page Designer + process



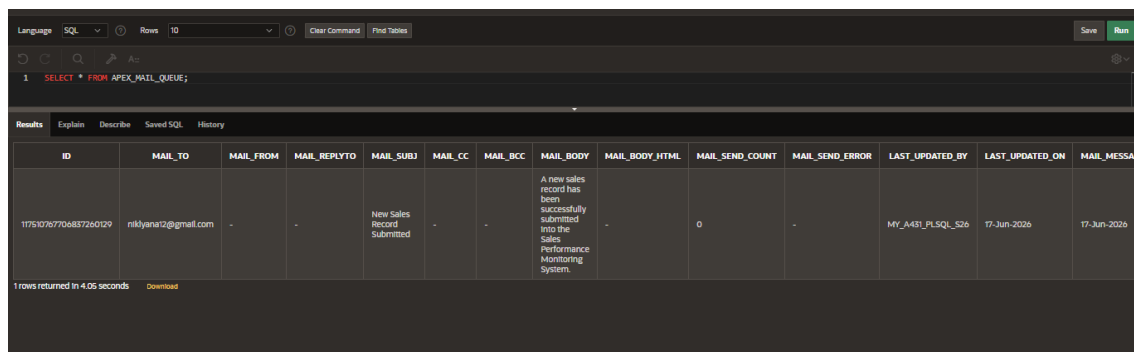
- Add process (type = send email) setting



iv. Row Created



v. Configuration E-mail (queue Email)



Language: SQL Rows: 10 Clear Command Find Tables Save Run

1 SELECT * FROM APEX_MAIL_QUEUE;

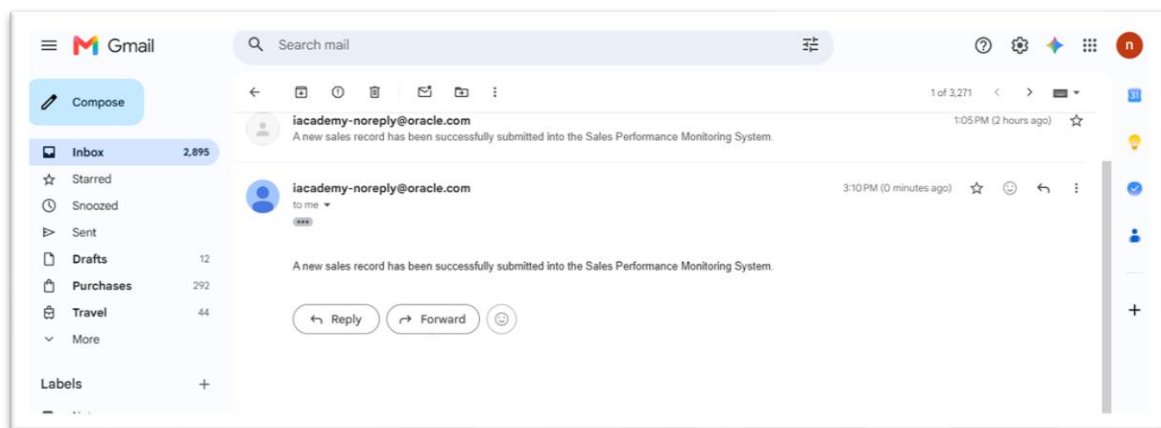
Results Explain Describe Saved SQL History

ID	MAIL_TO	MAIL_FROM	MAIL_REPLYTO	MAIL_SUBJ	MAIL_CC	MAIL_BCC	MAIL_BODY	MAIL_BODY_HTML	MAIL_SEND_COUNT	MAIL_SEND_ERROR	LAST_UPDATED_BY	LAST_UPDATED_ON	MAIL_MESSAGE
11751076710683726129	niklyana12@gmail.com	-	-	New Sales Record Submitted	-	-	A new sales record has been successfully submitted into the sales Performance Monitoring System.	-	0	-	Mf_A431_PLSQL_526	17-Jun-2026	17-Jun-2026

1 rows returned in 4.05 seconds Download

The email queue was checked “select * from APEX_EMAIL_QUEUE;” to verify that the notification email was successfully generated and queued for delivery.

vi. Received E-mail



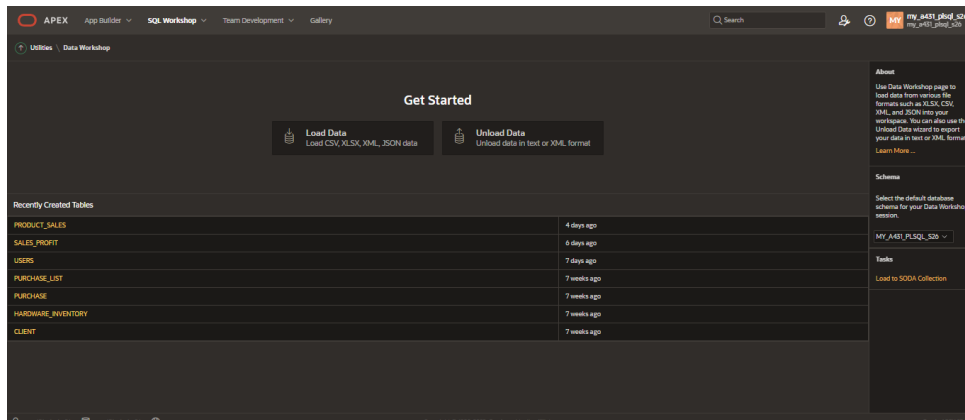
The notification email was successfully received to niklyana12@gmail.com , confirming that the Send Email process works correctly.

Summary task 3:

In this task, a sales submission form was created in Oracle APEX. A Send Email process was configured to send notification emails automatically whenever a new sales record was submitted.

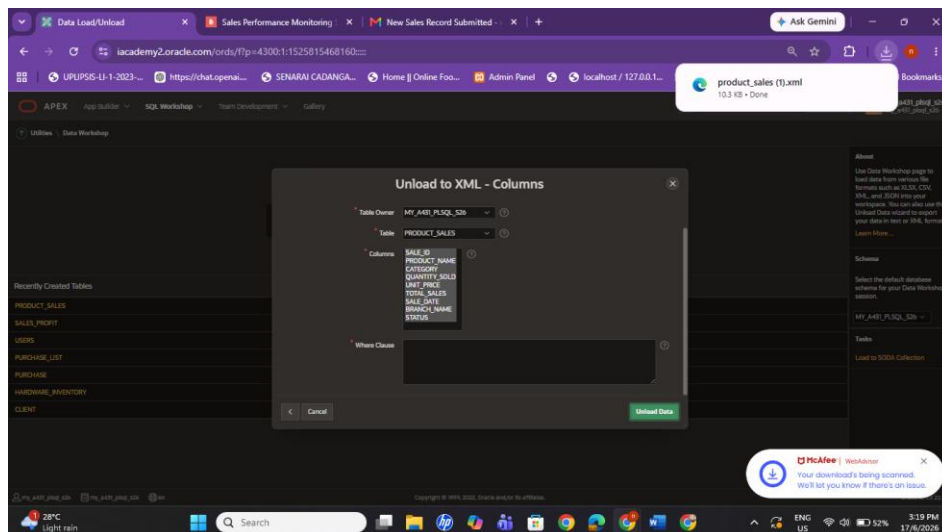
Task 4 : Migrate Database Objects and Data

PART A - Export PRODUCT_SALES Table

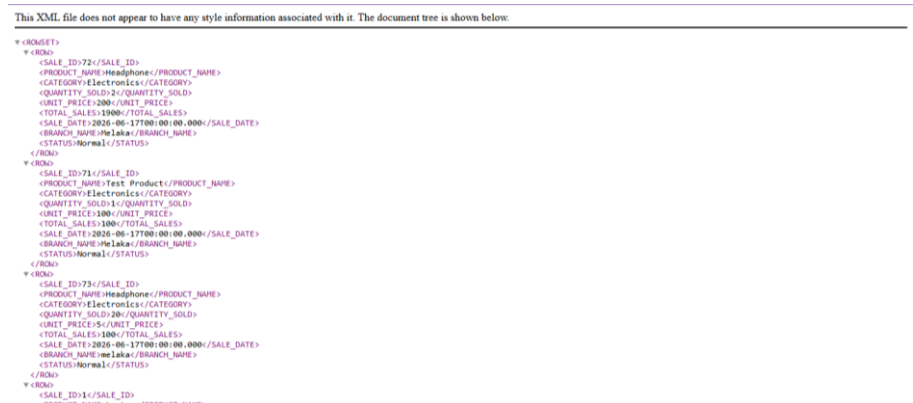


a. Unload Data

First the table data was exported using the Unload Data feature and saved in XML format.



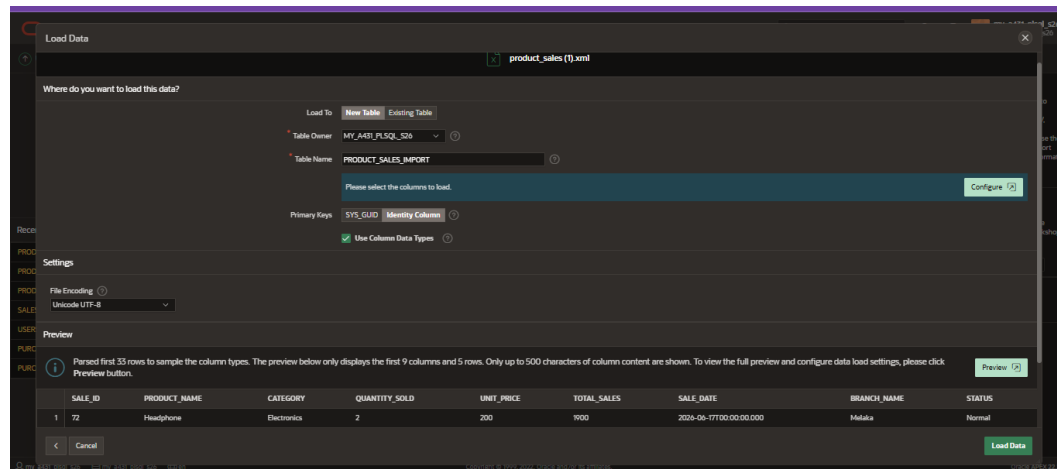
result (xml.)



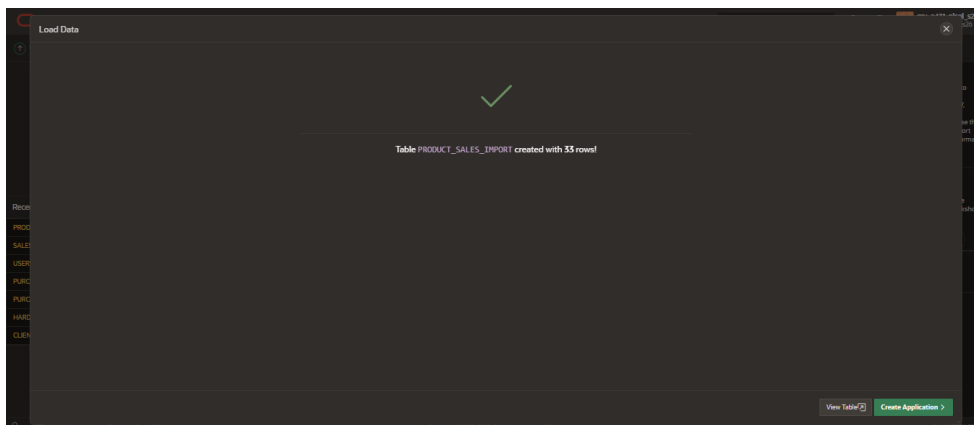
b. Load data (.CSV)

Second , using load data the exported data was imported from the export (PRODUCT_SALES) XML format to insert the new data or schema like below :

- Rename new table



- Result after load data :



c. Verify Imported Records

PRODUCT_SALES_IMPORT						
Column Name	Data Type	Nullable	Default	Primary Key		
ID	NUMBER	No	"MY_A431_PLSQL_S26"."SEQ\$S_0100700".nextval	1		
SALE_ID	NUMBER	Yes	-			
PRODUCT_NAME	VARCHAR2(100)	Yes	-			
CATEGORY	VARCHAR2(50)	Yes	-			
QUANTITY_SOLD	NUMBER	Yes	-			
UNIT_PRICE	NUMBER	Yes	-			
TOTAL_SALES	NUMBER	Yes	-			
SALE_DATE	TIMESTAMP(9)	Yes	-			
BRANCH_NAME	VARCHAR2(100)	Yes	-			
STATUS	VARCHAR2(50)	Yes	-			

The imported records were verified to ensure that the data migration process was completed successfully.

Summary task 4 :

In this task, the PRODUCT SALES table and its data were exported and imported into another environment. The migration process was successfully completed, and the imported records were verified.

Conclusion :

The Sales Performance Monitoring System was successfully developed using Oracle APEX. The system supports sales reporting, data visualisation, automation, email notification, and database migration features to improve sales monitoring and management efficiency.